

While Loop in C Programming

The **while loop** in C is an entry-controlled loop that repeatedly executes a block of code as long as a given condition is true.

The condition is checked **before** the loop body runs. If the condition is false initially, the loop will not execute even once.

Syntax

```
while (condition) {  
    // statements to be executed  
}
```

How it Works

- **Step 1:** The condition is evaluated first.
- **Step 2:** If the condition is true (non-zero), the loop body executes.
- **Step 3:** After execution, control goes back to recheck the condition.
- **Step 4:** The loop continues until the condition becomes false (0).

Example Implementation

```
#include <stdio.h>
```

```
int main() {  
    int i = 1; // Initialization  
  
    while (i ≤ 5) { // Condition  
        printf("%d ", i);  
        i++; // Update  
    }  
  
    return 0;  
}
```

Output: 1 2 3 4 5

Key Points to Remember

- It is an **Entry-controlled loop**.
- The loop variable must be initialized before the loop begins.
- The condition must eventually become false to avoid an **infinite loop**.
- Ideal for situations where the number of iterations is not known in advance.

Infinite While Loop

If the condition always remains true, the loop will never stop:

```
while (1) {  
    printf("Hello Forever  
");  
}
```

Common Uses

- Reading user input until a specific exit keyword or condition is met.
- Developing menu-driven programs.
- Repeating tasks where the termination depends on a dynamic runtime factor.